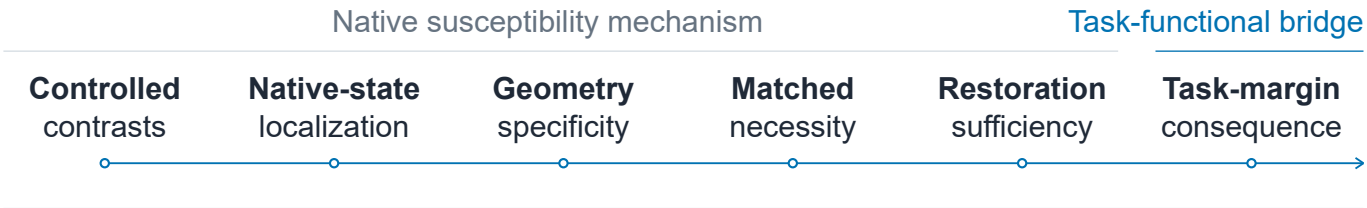


A Deep causal establishment at Mamba-130M



B Five scales: causal role recurs; geometry is only partially conserved

130M	370M	790M	1.4B	2.8B
P3 / P5	P3 / P5	P2 / P5	P5 / P4	P3 / P5
• Supported	• Supported	• Supported	• Supported	• Supported

Selected / control are scale-local ranks; equal rank numbers do not imply semantic homology.

C Objective-conditioned functional readout

